

Frontend Lead & Farmer Experience Engineer

Focus: Farmer Slot Booking Portal, Live 5-Stage Tracker & Digital Receipts

Executive Summary & Core Responsibility:

Crafted the modern, farmer-centric UI/UX, responsive booking portal, dynamic progress tracker, and official e-Receipt vouchers.

1. Technology Stack & Architectural Components

Domain / Layer	Specific Technologies, Libraries & Frameworks Used
Core Frontend	HTML5, Modern JavaScript (ES6+ async/await, Fetch API, DOM APIs)
Styling & Design	Tailwind CSS (Custom Kisan Emerald theme), CSS Grid, Responsive Flexbox
Client State	Local Storage persistence, dynamic DOM mutation, LiveSync WebSocket client
Audio/Visual Feedback	Web Audio API (Synthesized chime frequencies), Dynamic CSS Pulse Rings
Print Engine	Print CSS Media Queries for verified Govt e-Receipt & Payment Voucher generation

2. Key Modules Developed & Production Deliverables

- **Farmer Slot Booking Interface (farmer_book.html & farmer.js):** Built 3-step intuitive booking workflow with instant dynamic MSP payout calculator (Quantity x Rate).
- **Live 5-Stage Visual Tracker (farmer_track.html):** Created real-time progress bar, stage-specific active glow badges, vehicle-ahead counters, and dynamic wait-time indicators.
- **Official Verified e-Receipt / Voucher:** Designed printable and downloadable digital procurement voucher with official Govt headers, net quantity calculation, and Aadhaar DBT reference.
- **Accessibility '■■■■ ■■■■' (Simple Mode):** Implemented large-font high-contrast accessibility switch for elder farmers and low-vision users.

3. Direct Codebase Ownership & Modified Files

Source Files:

frontend/templates/farmer_book.html • frontend/templates/farmer_track.html • frontend/static/js/farmer.js • frontend/static/css/styles.css

4. SIH Jury Defense & Live Evaluation Points (What to Speak Tomorrow)

- **Presentation Pitch Point:** Demonstrate how a farmer books a slot in under 30 seconds with instant feedback.
- **Presentation Pitch Point:** Showcase the 5-stage visual card progression updating live without page reloads.
- **Presentation Pitch Point:** Highlight the official printable e-Receipt voucher with digitized verification stamp.